

ST MICHAEL LOEBEN, Austria

WMO: 111740

Lat: 47.333N

Lon: 15.000E

Elev: 1877

StdP: 13.73

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.1	11.4	4.2	7.3	9.5	8.1	8.9	14.2	15.8	40.5	13.9	39.9	1.3	250	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	85.6	67.5	82.1	66.2	78.8	64.6	69.2	81.7	67.5	79.0	65.9	76.0	5.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	99.8	72.9	63.6	94.4	71.5	62.3	90.1	70.0	34.4	81.8	33.0	79.1	31.7	76.3	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.4	11.2	9.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			1.9	91.3	5.3	3.1	-1.9	93.6	-5.1	95.4	-8.0	97.1	-11.9	99.4
			1.4	72.5	5.1	2.1	-2.3	74.0	-5.3	75.3	-8.1	76.5	-11.8	78.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	27.5	32.7	39.9	48.0	56.7	62.5	65.6	64.9	57.3	48.6	38.8	29.5
	DBStd	14.84	7.45	7.28	7.25	6.70	5.98	6.30	5.60	5.20	5.92	6.88	7.18	6.94
	HDD50	2717	698	485	316	114	10	1	0	0	9	109	340	635
	HDD65	6495	1163	905	777	510	263	120	62	64	238	507	786	1100
	CDD50	1898	0	0	4	54	219	376	484	462	228	67	4	0
	CDD65	199	0	0	0	0	7	44	80	61	7	0	0	0
	CDH74	1981	0	0	0	10	147	471	728	558	66	1	0	0
	CDH80	510	0	0	0	0	24	118	214	149	5	0	0	0
(14) Wind	WSAvg	3.0	2.9	3.2	3.7	3.5	3.3	3.0	3.2	2.6	2.3	2.5	2.9	2.6
(15) Precipitation	PrecAvg	40.2	1.8	1.8	2.3	2.5	4.0	5.3	5.8	5.3	3.9	3.1	2.5	1.9
	PrecMax	49.2	4.2	3.7	6.1	5.0	7.0	7.8	12.7	8.7	5.9	6.2	4.7	3.8
	PrecMin	30.2	0.4	0.3	0.7	0.4	2.1	3.0	1.7	2.5	2.0	0.7	0.0	0.2
	PrecStd	4.5	1.1	0.9	1.1	1.0	1.3	1.3	2.2	1.7	1.1	1.3	1.0	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	56.6	65.5	74.9	83.2	87.8	90.1	89.6	80.3	71.4	63.7	48.3
		MCWB	42.2	45.8	51.3	58.0	64.0	68.3	69.3	68.4	65.0	59.9	53.7	42.3
	2%	DB	43.7	51.0	60.6	69.9	77.9	83.7	85.6	84.3	75.4	66.2	55.2	42.9
		MCWB	39.4	42.5	48.7	54.8	61.8	66.4	68.3	66.8	62.6	57.2	49.8	39.7
	5%	DB	39.9	46.6	56.2	65.9	74.2	79.9	81.8	80.1	71.7	62.1	51.6	39.8
		MCWB	36.4	40.2	46.8	53.0	59.9	64.9	66.7	65.5	60.6	54.9	47.3	37.4
	10%	DB	36.9	43.0	52.0	61.5	70.1	75.9	77.8	76.5	68.0	58.6	48.3	37.7
		MCWB	34.3	37.9	44.8	50.6	57.9	63.1	65.0	64.2	59.2	53.2	45.1	35.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	46.6	53.0	59.0	65.5	70.5	71.7	70.5	66.3	61.4	53.7	43.9
		MCDB	48.2	56.1	62.4	73.1	78.8	82.7	84.6	82.9	76.8	68.6	61.4	47.1
	2%	WB	39.5	43.2	49.4	56.2	62.9	68.3	69.5	68.4	64.1	58.6	50.6	40.1
		MCDB	43.3	50.0	58.8	67.1	75.0	80.2	82.5	80.5	72.8	64.7	54.8	42.3
	5%	WB	36.7	40.8	47.4	53.9	61.1	66.3	67.6	66.6	62.0	56.3	48.2	37.8
		MCDB	39.3	45.8	55.4	63.5	71.9	77.1	79.2	77.3	69.2	61.0	51.1	39.4
	10%	WB	34.7	38.2	45.1	51.6	59.0	64.3	66.0	65.0	60.1	53.8	45.7	36.0
		MCDB	36.4	42.4	51.1	59.6	68.3	73.5	75.6	74.3	66.2	57.5	47.9	37.3
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	16.3	18.2	20.1	20.9	20.0	20.3	19.9	18.7	16.6	11.7	10.3
		MCDBR	14.6	22.4	27.2	29.2	29.2	28.3	27.6	27.8	24.4	21.2	16.0	11.9
		MCWBR	10.1	14.2	15.9	14.9	13.7	12.3	11.4	11.6	12.1	12.7	10.6	8.9
	5% WB	MCDBR	13.4	20.8	25.6	26.0	26.8	25.2	25.2	24.8	22.1	19.5	14.4	10.6
		MCWBR	9.6	13.6	15.3	13.6	12.9	11.6	11.0	11.0	11.4	12.0	9.9	8.1
(40) Clear-Sky Solar Irradiance	taub		0.290	0.319	0.362	0.407	0.414	0.420	0.422	0.419	0.387	0.357	0.325	0.291
	taud		2.404	2.342	2.289	2.226	2.258	2.279	2.275	2.301	2.362	2.422	2.463	2.462
	Ebn at Noon		253	267	271	268	270	268	266	262	259	250	238	240
	Edh at Noon		22	29	35	41	42	41	41	38	33	26	21	18
(44) All-Sky Solar Radiation	RadAvg		384	657	991	1312	1532	1681	1678	1497	1082	740	414	318
	RadStd		43	83	113	139	169	127	147	171	138	72	60	46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (109 neighbors)	+0.99	N/A	N/A	+1.60	+0.85	+0.72	-194	-261	+166	+99

Nomenclature: See separate page